

b-1.

$$k_{sp} = \sqrt{\frac{10^{0.105} - 1}{10^{0.105} - 1}} = 17.794$$

$$\lambda_{sp} = \frac{2\pi f_s}{2\pi f_p} = 2$$

$$N = \left\lceil \frac{19k_{sp}}{19\lambda_{sp}} \right\rceil = 5$$

12)

求 \$f_c\$:

$$G(p) = \frac{1}{p^5 + 3.2361p^4 + 5.2361p^3 + 5.2361p^2 + 3.2361p + 1}$$

13) 带宽 \$B_p = 3\$ dB 则 \$f_{oc} = f_p = 2\pi f_p = 2\pi \times 6 \times 10^3\$ rad/s

$$H(s) = H(p) \Big|_{p = \frac{s}{\omega_c}}$$

$$= \frac{\omega_c^5}{s^5 + 3.2361\omega_c s^4 + 5.2361\omega_c^2 s^3 + 5.2361\omega_c^3 s^2 + 3.2361\omega_c^4 s + \omega_c^5}$$

b-5 m

① 冲激响应的求法

$$H(s) = \frac{1}{s^2 + s + 1}$$

$$s_1 = -\frac{1}{2} + j\frac{\sqrt{3}}{2}, s_2 = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$$

$$H(s) = \frac{-j\frac{\sqrt{3}}{2}}{s - (-\frac{1}{2} + j\frac{\sqrt{3}}{2})} + \frac{j\frac{\sqrt{3}}{2}}{s - (-\frac{1}{2} - j\frac{\sqrt{3}}{2})}$$

$$H(z) = \frac{-j\frac{\sqrt{3}}{2}}{1 - e^{(-\frac{1}{2} + j\frac{\sqrt{3}}{2})T} z^{-1}} + \frac{j\frac{\sqrt{3}}{2}}{1 - e^{(-\frac{1}{2} - j\frac{\sqrt{3}}{2})T} z^{-1}}$$

\$T=2\$

$$H(z) = \frac{\frac{\sqrt{3}}{3}}{1 - 2e^{-1}\cos\frac{\sqrt{3}}{2}z^{-1} + e^{-2}z^{-2}}$$

$$H(s) = \frac{\frac{\sqrt{3}}{2}}{(s + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \cdot \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}}$$

求 \$a, b\$ 得 \$a = \frac{1}{2}, b = \frac{\sqrt{3}}{2}\$

$$H(z) = \frac{\frac{\sqrt{3}}{3}}{1 - 2e^{-1}\cos\frac{\sqrt{3}}{2}z^{-1} + e^{-2}z^{-2}} \Big|_{T=2}$$

$$= \frac{\frac{\sqrt{3}}{3}}{1 - 2e^{-1}\cos\frac{\sqrt{3}}{2}z^{-1} + e^{-2}z^{-2}}$$

双线性变换法:

$$H(z) = H(s) \Big|_{s = \frac{1-z^{-1}}{1+z^{-1}}}, T=2 = \frac{1}{\left(\frac{1-z^{-1}}{1+z^{-1}}\right)^2 + C \frac{1-z^{-1}}{1+z^{-1}} + 1}$$

$$= \frac{1+z^{-1}+z^{-2}}{3+z^{-2}}$$

(2)

冲激响应不变法:

$$H(s) = \frac{1}{s+1/2} + \frac{-1}{s+1}$$

$$H(z) = \frac{1}{1-e^{-1/2}z^{-1}} + \frac{-1}{1-e^{-1}z^{-1}} \Big|_{T=2}$$

$$= \frac{1}{1-e^{-1}z^{-1}} - \frac{1}{1-e^{-2}z^{-1}}$$

双线性变换法:

$$H(z) = H(s) \Big|_{s = \frac{1-z^{-1}}{1+z^{-1}}} = \frac{1+z^{-1}+z^{-2}}{6-2z^{-1}}$$

6-9.

已知 $W_p = 0.2\pi$, $A_p = 10\text{dB}$

$W_s = 0.3\pi$, $A_s = 10\text{dB}$

$\omega_p = \frac{W_p}{T} = 200\pi \text{ rad/s}$, $\omega_s = \frac{W_s}{T} = 300\pi \text{ rad/s}$

$k_{sp} = \sqrt{\frac{10^{0.1A_p}-1}{10^{0.1A_s}-1}} = 0.1676$

$\lambda_{sp} = \frac{\omega_s}{\omega_p} = \frac{300\pi}{200\pi} = 1.5$

$N = -\frac{\lg k_{sp}}{\lg \lambda_{sp}} = 4.376$ 向上取整 $N=5$

查表

$$G_p(s) = \frac{1}{\prod_{k=0}^N (p - p_k)}$$

$p_0 = -0.3090 + j0.9511 = p_4^*$, $p_2 = -1$

$p_1 = -0.8090 + j0.5818 = p_3^*$

$\text{PM } \omega_c = \omega_s (10^{0.1A_s}-1)^{-\frac{1}{2N}}$

$= 756.5661 \text{ rad/s}$

$H(s) = G_p(s) \Big|_{p = \frac{s}{\omega_c}}$

$\text{PM } H(z) = \frac{4}{k_{sp}} \frac{B_k}{1-e^{j\omega_k} z^{-1}}$

6-10

$$^{11} \quad \omega_p = \frac{2}{T} \tan \frac{\omega T}{2} = 0.6498394 \text{ rad/s}$$

$$\omega_s = \frac{2}{T} \tan \frac{\omega_s T}{2} = 1.0190509 \text{ rad/s}$$

$$k_{sp} = 0.1676$$

$$x_{sp} = \frac{\omega_s}{\omega_p} = 1.5682$$

$$N = -\frac{\lg k_{sp}}{\lg x_{sp}} = 3.9425, \quad N=4.$$

查表 $a_{cp} = \frac{1}{s^4 + 2.6131s^3 + 1.4142s^2 + 2.6131s + 1}$

$$\omega_c = \omega_p (10^{0.16P} - 1)^{-\frac{1}{2N}} = 0.7743 \text{ rad/s}$$

$$H_a(s) = G(p)|_p = \frac{s}{\omega_c} = \frac{0.3595}{s^4 + 2.0234s^3 + 2.0470s^2 + 1.2131s + 0.3595}$$

$$H(z) = |H_a(s)|_{s=\frac{2}{1+z^{-1}}} \quad T=1$$

